

WASP-12b

R-2069

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_150126 · created 2026-08-09T23:04:28+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457048.71174 +/- 0.00013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.044
Transit depth (Rp/Rs)^2	2.4 +/- 1.35 [%]
Orbital Inclination (inc)	85.69 +/- 2.91
Airmass coefficient 1 (a1)	1.39 +/- 0.17
Airmass coefficient 2 (a2)	-0.29 +/- 0.21
Scatter in the residuals of the lightcurve fit is	12.33 %
Best Comparison Star	#3 - [527, 93]
Optimal Aperture	2.66
Optimal Annulus	7.28
Transit Duration (day)	0.1321 +/- 0.0067

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.155 ± 0.044 vs 0.1178 (deviation 0.79σ)
Duration fit vs published	0.1321 d vs 0.125 d (deviation 0.99σ)
Mid-time O–C	0.2 min
Depth SNR	3.3
Residual scatter	12.33 %

Light curve

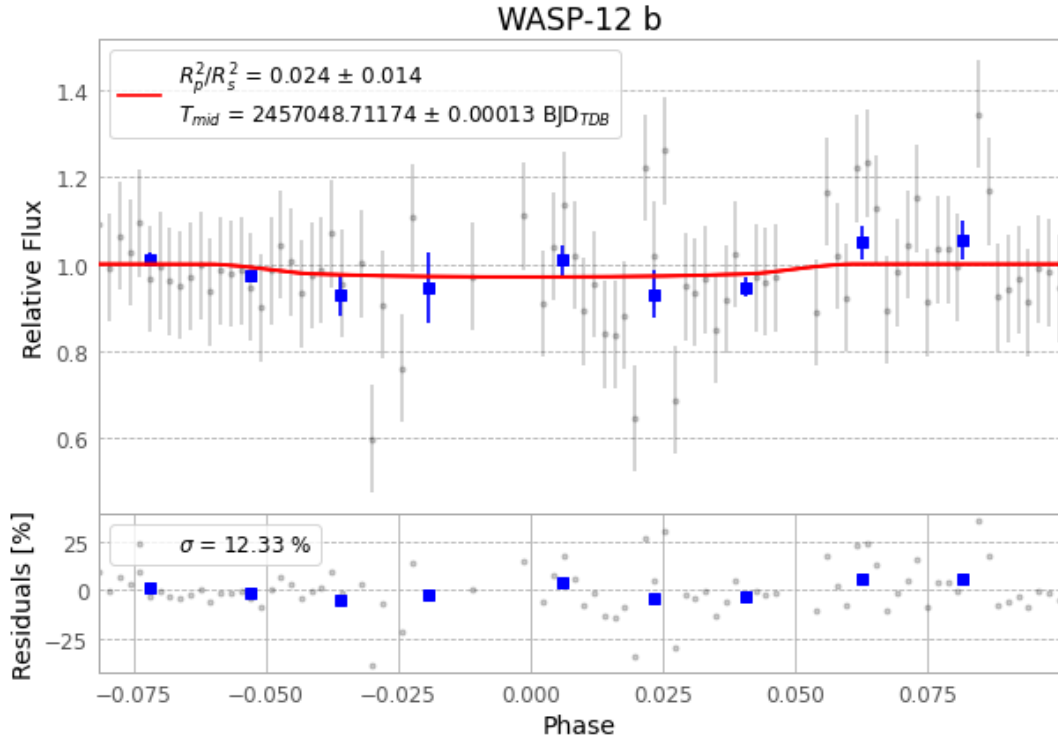


Figure 1 — FinalLightCurve_WASP-12 b_2015-01-25.png (SHA-256 2d585502674fdf4b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [341, 278], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 3d68bd8ca7e386d0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

98 FITS frames (64 MB), WASP-12150126024812.FITS → WASP-12150126073912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-150126.log (f2689a37590a...), FinalLightCurve_WASP-12 b_2015-01-25.png (2d585502674f...), FinalLightCurve_WASP-12 b_2015-01-25.csv (ae184d4615aa...)

Compute resources

Wall time 248.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-09 23:04Z.